

HACHIOJIMA, Japan

WMO: 476780

Lat: 33.1217N

Lon: 139.7792E

Elev: 501

StdP: 14.43

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.8	42.4	24.7	19.1	43.7	27.2	21.5	44.9	33.6	54.4	30.0	52.3	12.9	280	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	86.1	79.0	84.8	78.5	83.6	78.0	80.4	83.8	79.7	82.9	79.0	82.0	9.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.4	155.6	82.2	78.7	151.8	81.7	78.0	148.3	81.1	44.5	84.0	43.7	83.1	43.0	82.3	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.3	25.6	22.8	36.1	88.6	2.5	1.1	34.3	89.4	32.8	90.0	31.4	90.7	29.6	91.4		
			32.0	81.8	1.9	0.9	30.7	82.4	29.6	83.0	28.5	83.5	27.2	84.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.1	50.4	51.0	54.7	60.9	66.9	71.6	78.2	80.4	77.1	70.8	63.1	55.0
	DBStd	11.02	4.42	4.85	4.82	4.29	3.16	3.25	2.78	2.06	3.46	3.70	4.38	4.85
	HDD50	115	48	42	13	0	0	0	0	0	0	0	0	12
	HDD65	1715	453	393	319	132	17	1	0	0	2	86	312	
	CDD50	5620	60	69	160	327	524	647	876	943	812	644	392	166
	CDD65	1744	0	0	0	9	76	198	411	478	362	181	28	1
	CDH74	10758	0	0	0	1	39	393	3030	4423	2443	418	11	0
	CDH80	1707	0	0	0	0	0	9	415	915	356	12	0	0
Wind	WSAvg	11.3	12.9	12.8	12.7	11.6	10.4	10.4	9.7	8.8	10.7	12.4	11.4	12.0
Precipitation	PrecAvg	124.4	7.3	7.6	10.8	9.1	10.0	14.4	8.1	7.6	13.5	17.8	10.7	7.3
	PrecMax	181.4	20.2	19.3	24.6	21.0	33.6	28.9	28.0	24.9	32.3	41.5	26.1	17.2
	PrecMin	86.3	2.2	1.3	4.4	3.0	3.8	2.8	0.1	1.1	2.2	5.4	3.3	0.9
	PrecStd	18.6	4.2	3.7	4.3	3.8	4.5	5.6	6.0	4.9	7.0	8.2	4.9	3.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.9	65.8	67.5	72.5	77.0	81.3	86.6	88.2	86.7	81.5	75.5	70.5
		MCWB	61.1	62.7	61.2	65.3	69.2	76.3	79.3	79.6	78.5	75.4	71.2	67.5
	2%	DB	62.4	63.5	65.8	70.4	75.1	78.8	85.1	86.7	85.0	79.6	73.0	66.9
		MCWB	58.0	59.9	61.3	64.7	68.4	74.8	78.9	79.3	78.1	74.0	67.4	62.2
	5%	DB	59.6	61.3	64.3	68.9	73.6	77.4	83.8	85.4	83.6	78.0	71.4	64.3
		MCWB	54.6	56.9	59.7	63.7	67.5	74.2	78.4	78.7	77.4	72.9	66.0	59.1
	10%	DB	57.2	58.9	62.7	67.4	72.2	76.3	82.4	84.2	82.1	76.4	69.7	62.2
		MCWB	51.8	53.9	57.8	62.4	66.9	73.5	77.8	78.3	76.8	71.8	64.7	57.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.7	64.1	65.4	69.0	73.5	77.5	80.9	81.8	80.6	77.4	72.9	67.5
		MCDB	64.3	65.2	66.0	70.6	75.0	79.3	84.4	85.2	84.2	79.5	74.7	69.4
	2%	WB	59.0	61.2	63.1	67.3	71.5	76.4	79.9	80.8	79.4	75.9	69.7	63.5
		MCDB	61.6	63.0	64.5	68.8	72.8	77.6	83.1	84.2	82.6	78.1	71.6	65.7
	5%	WB	55.5	57.9	60.9	65.5	70.1	75.5	79.1	80.0	78.6	74.5	67.7	60.5
		MCDB	58.7	60.4	63.5	67.5	71.7	76.7	82.0	83.3	81.7	76.9	70.2	63.1
	10%	WB	52.7	54.9	58.6	63.6	68.7	74.4	78.4	79.3	77.9	73.0	65.5	57.9
		MCDB	56.4	58.1	62.0	66.4	70.8	75.7	81.1	82.6	81.0	75.5	68.6	61.3
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.1	9.8	9.4	8.8	6.9	6.9	7.9	8.0	7.8	8.5	8.3
		MCDBR	11.3	11.7	11.0	10.2	10.1	7.6	8.5	9.1	9.2	8.9	10.0	10.6
		MCWBR	10.9	11.8	10.1	7.7	7.1	5.0	4.6	4.8	5.1	6.2	8.2	9.9
	5% WB	MCDBR	11.0	11.3	10.3	8.3	7.3	6.1	7.6	8.2	7.8	8.7	9.7	
		MCWBR	11.8	13.0	11.6	8.8	7.1	4.6	4.6	4.5	5.1	6.8	9.2	10.9
Clear-Sky Solar Irradiance	taub	0.395	0.439	0.487	0.513	0.502	0.503	0.498	0.473	0.456	0.454	0.420	0.396	
	taud	2.232	2.113	1.991	1.977	2.085	2.184	2.230	2.313	2.334	2.249	2.248	2.232	
	Ebn at Noon	253	253	250	250	254	253	253	258	258	246	243	245	
	Edh at Noon	36	44	54	57	52	47	45	41	38	38	35	34	
All-Sky Solar Radiation	RadAvg	712	899	1210	1472	1633	1452	1842	1880	1453	1022	779	674	
	RadStd	69	102	88	152	163	173	216	113	114	91	58	60	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	-0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-203	-128
(47)	Regional (3 neighbors)	N/A	N/A	+1.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page